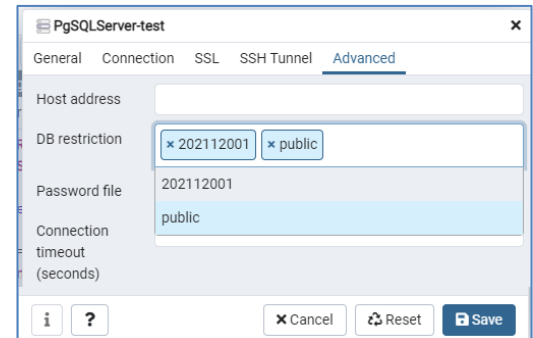


Lab04 Querying operations on relations

IT214 Database Management System, Winter'2026; pm_jat @ daiict

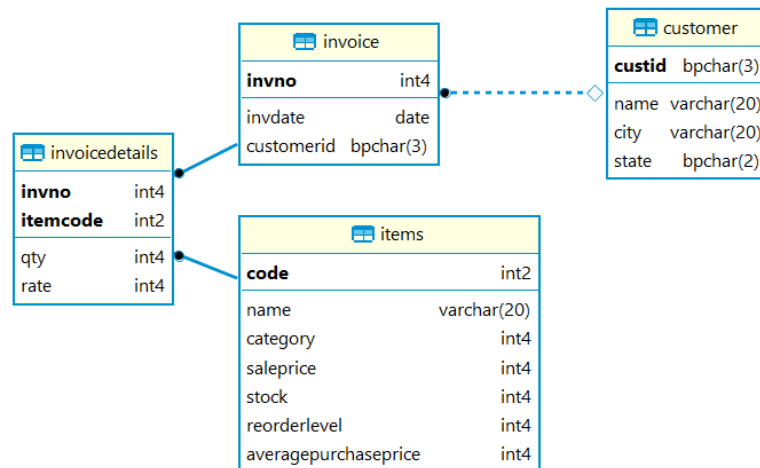
In this lab, you learn and practice querying databases using Relational Algebra and SQL. For practice purposes, you are given access to a database named as **public** (on the server 10.100.71.21). Where various schemas are populated with some sample data that you can use for practice purposes.

Before accessing, add this database to DB Restriction in your PgAdmin client, while you are disconnected from the server; as follows: right-click on server-name; click on properties; go to the advanced tab; add public database; typically, as shown below, and save it.



In this lab you perform querying operations on two database schemas, SALES and ACAD. Both are available in the **public** database. You have only read permissions on these databases. Your goal here is to write relational algebra expressions and SQL statements for answering the given queries. [Algebraic solutions can be handwritten and scanned for submission purpose]

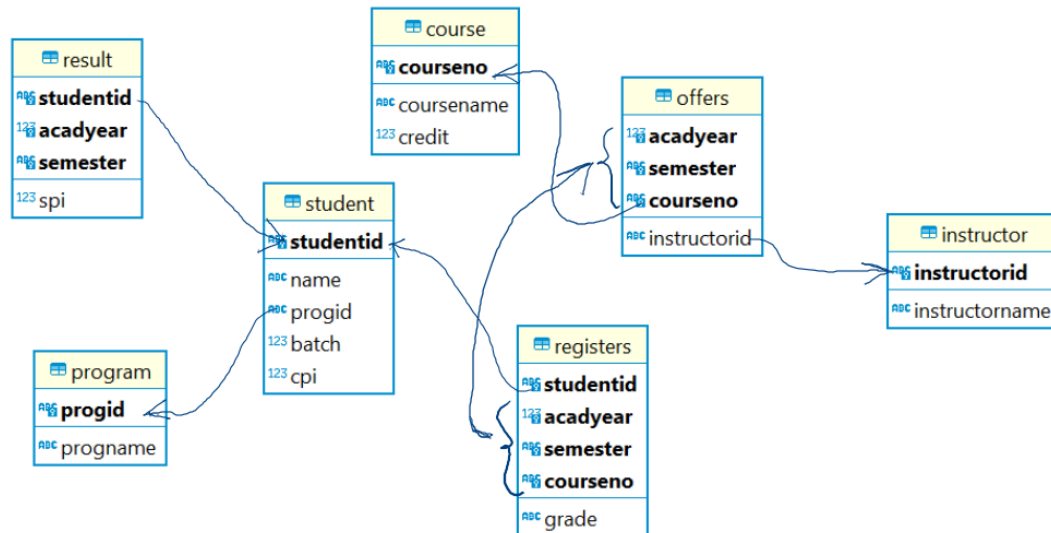
Queries on SALES schema:



1. Give the details of items having the price > 1000 and belonging to category 5;
2. List the Invoices (number, date) of a customer id = 'C05'
3. List all invoices (customer id, date, inv no, qty) that have item code = 1103
4. List all items (inv no, item code, qty) that customer 'C05' has bought
5. List all customers (customer id, name, city) that have bought item 1101
6. List all items (item code, item name) sold on 2011-08-23
7. List all items (item code, item name) sold in the month '2011-07' from category=3
8. List customers (customer id, name, city) who are from Gujarat and have bought item items from category=3

[Submission Required: Query Expressions in Relational Algebra and SQL]

Queries on ACAD schema:



[Note: Attribute “acadyear” appearing in multiple relations stores the beginning year of the academic year. For example, academic year 2024-25, would be recorded as 2024 in this attribute]

9. List Faculty (ID, Name) for all courses of Autumn'2010
10. List Students (ID, Name) from batch 2008 registered for course MT101 in Autumn'2008.
11. List Students (ID, Name) from B Tech (progid='01') batch 2008 registered for course MT101 in Autumn'2008.
12. List courses that were offered but no students were registered in those offerings
13. List students (id, name, courseno, grade) who got 'F' grade in Autumn'2008
14. List Courses (Course No, Title, Credits) offered by Instructor (ID='PMJ') in semester Winter'2010.
15. Produce grade-sheet for student '200711002' for Academic year 2008-09; i.e.
List Course No, Course Name, Course Credit, Grade
16. List ID of students who have taken both courses - 'MT101' AND 'MT104'.

[Submission Required: Query Expressions in Relational Algebra and SQL]

Submission Instructions: Submit everything in a single PDF file.